

Technical Architecture

Bounteous Proprietary & Confidential



Project Name: AI-Powered Knowledge Assistant

Team Name: Hello World

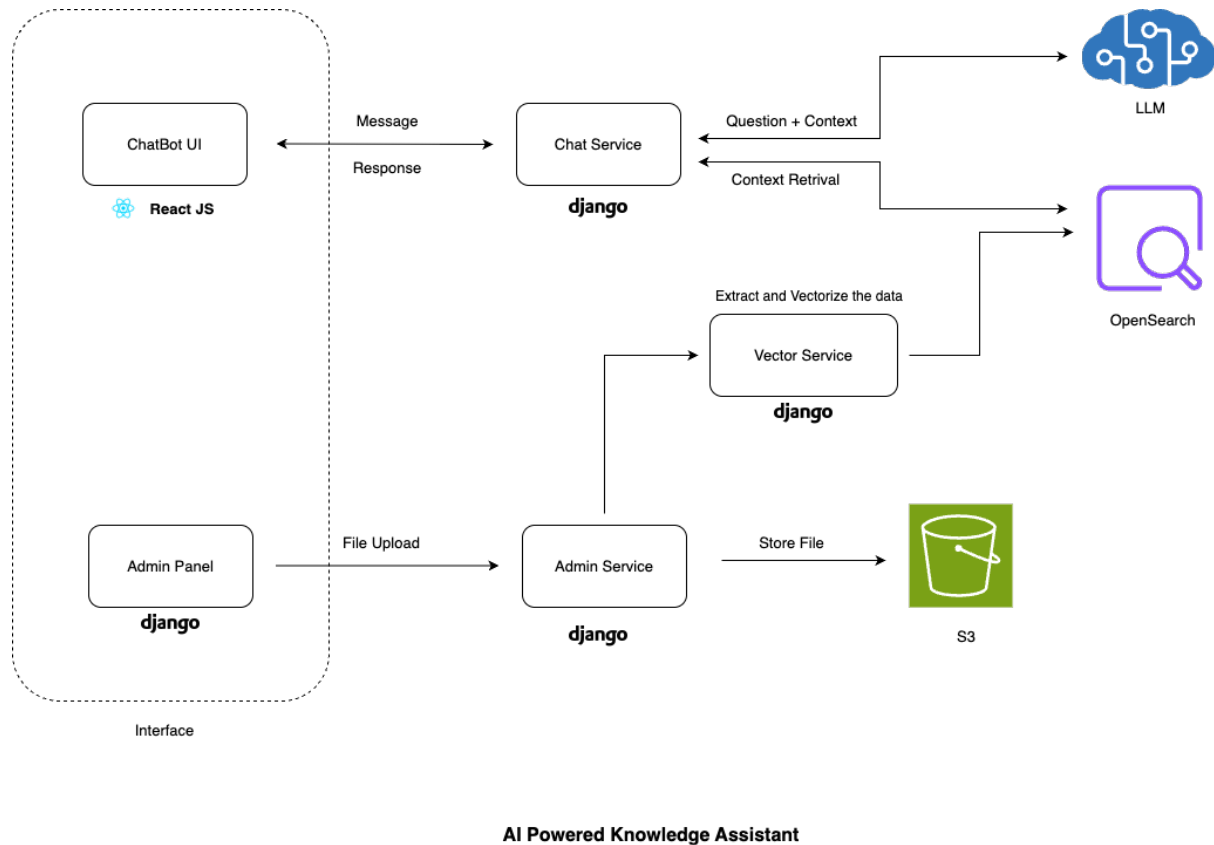
Version: 1.0

Date: 04/06/2025

Author(s): Yasmeen Mohammed Rafi , Mugunth Chandirasekaran, Karthick Ramesh, Nikhil Anurag, Aseem Narayan Pandey

Document Status:

Technical Architecture



System Architecture Patterns

- **Architectural Style:** Microservices-inspired Django monolith
- **Design Patterns:** Modular MVC + DAO pattern for DB access
- **Communication Protocols:** REST APIs

Infrastructure Architecture

- **Deployment Environment:** AWS EC2 or containerized setup
- **Container Strategy:** Docker with future Kubernetes compatibility
- **Load Balancing:** AWS ALB or reverse proxy using NGINX (Planned)
- **Monitoring & Logging:** CloudWatch / Splunk planned (Planned)

Network Architecture

- **Network Topology:** VPC with private subnets for services
- **API Gateway:** Django REST endpoint behind ALB
- **Service Mesh:** Not required in current MVP

- **Security Boundaries:** IAM, HTTPS, VPC security groups

Data Architecture

- **Data Sources:** PDFs/Document uploads
- **Data Pipeline:** Upload → Parse → Chunk → Embed → Index
- **Data Warehouse/Lake:** Not applicable
- **Real-time Processing:** Query-time vector search
- **Data Governance:**
 - Document metadata tracked
 - Timestamp and filename stored per chunk

Disaster Recovery & Business Continuity

- **Backup Strategy:**
 - S3 object versioning and lifecycle rules
 - OpenSearch snapshot repo setup
- **Recovery Procedures:** Manual re-indexing from saved documents
- **Failover Mechanisms:** Pilot Light approach
- **Business Continuity Plan:** Scheduled backups + CloudWatch alerts + replication (optional)